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A rapid and sensitive method for the determination of vitamin E and carotenoids in vegetables and fruits using deep eutectic solvent-based dispersive liquid-liquid microextraction

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ABSTRACT

Conventional extraction techniques for fat-soluble nutrient analysis in foods may present certain limitations, including variable or limited efficiency, high solvent consumption, lengthy procedures, and potential environmental and health concerns. This study evaluates deep eutectic solvents (DESs) as a green and efficient alternative for extracting vitamin E and carotenoids from vegetables and fruits. Conductor-like screening model for real solvents (COSMO-RS) simulation and experimental screening identified the menthol-thymol (1:2) DES as optimal due to the balanced distribution of polar and nonpolar sites within its components. A novel dispersive liquid-liquid microextraction method based on this DES (DES-DLLME) was developed and integrated with HPLC for the simultaneous quantitation of vitamin E and carotenoids. The optimized method demonstrated excellent analytical performance with low detection limits (0.01–0.18 μg/mL), satisfactory recovery rates (80.04–111.95 %), and high reproducibility (relative standard deviation, RSD < 16.83 %). When applied to a range of Chinese food samples, the obtained nutrient profiles were largely consistent with USDA reference data, with observed variations likely due to factors such as geographic origin, cultivation practices, and post-harvest processing. This DES-based approach offers an efficient, reproducible, and environmentally friendly method for determining fat-soluble vitamins and carotenoids in complex food matrices, offering significant value for food quality assessment and nutritional research.

1. Introduction

Fruits and vegetables are primary dietary sources of bioactive compounds, particularly carotenoids, vitamin E isomers, and dietary fiber. These components collectively contribute to mitigating risks of chronic diseases, including cancer, cardiovascular disorders, and cataracts [1]. Epidemiological studies reveal that fresh produce contributes substantially to daily nutrient intake, accounting for approximately 25 % of dietary vitamin E (predominantly from apples, pears, tomatoes, and lettuce) and 70 % of provitamin A carotenoids (mainly sourced from carrots, tomatoes, spinach, and borage) [2]. However, the nutritional composition of these foods exhibits significant spatiotemporal variability due to regional seasonality, cultivation practices, and postharvest handling, necessitating robust food composition tables (FCTs) for accurate dietary assessment [3–5]. Currently, the FCT provided by the United States Department of Agriculture (USDA) [6] remains one of the most comprehensive databases, which incorporates detailed nutrient profiles, including changes induced by cooking. In contrast, the China Food Composition Tables (2018) quantifies total carotenoids via paper chromatography but lacks specificity for individual carotenoid isoforms or data on thermal processing effects - a methodological limitation that impedes cross-national data comparability. Preliminary analysis utilizing USDA data to estimate lutein and zeaxanthin intake in Chinese populations have revealed significant discrepancies, attributable to variations in food varieties, culinary practices, and analytical protocols [7]. This highlights the urgent need for region-specific compositional data, particularly given the scarcity of comparative studies on vitamin E and carotenoid profiles between Chinese and U.S. food systems.

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Accurate quantitation of vitamin E and carotenoids in plant matrices necessitates efficient pretreatment to eliminate matrix interferences such as sugars, lipids, and organic acids [8]. Due to the hydrophobic nature of carotenoids and vitamin E, traditional extraction methods typically rely on organic solvents including acetone [9], acetone-diethyl ether mixtures [10], hexane-ethanol-acetone-toluene blends [11], tetrahydrofuran [12], and tetrahydrofuran-methanol mixtures [13]. However, the varying polarity of different carotenoids complicates their simultaneous extraction, and most conventional organic solvents pose environmental and health risks [8]. Recent developments in extraction techniques, such as enzyme-assisted extraction (EAE) [14,15], microwave-assisted extraction (MAE) [16,17], ultrasound-assisted extraction (UAE) [18,19], supercritical fluid extraction (SFE) [20], and Pressurized liquid phase extraction (PLE) [21], aim to address these challenges [22]. For example, Sharma et al. [23]. demonstrated that ultrasonic and microwave-assisted extraction enhanced carotenoid extraction from SBT-P primarily through solvent optimization. Similarly, Hiranvarachat et al. [24]. improved β-carotene yield from carrot waste by applying intermittent microwave radiation. Jaeschke et al. [25]. achieved optimal carotenoid recovery from *Chlococcus lutealis* by using 75 % ethanol under 50 % ultrasonic intensity (total power 664 W) at 30 °C. Basegmez et al. [26]. compared SFE-CO_2, PLE, and EAE for tocopherol extraction from fruit pomace, finding PLE to yielded the highest output. Kalcal et al. [27]. employed PLE and EAE to extract α-, β-, γ-, and δ-tocopherols from sea buckthorn pomace and seeds, and further assessed the antioxidant activity of the extracted compounds. Despite these innovations, persistent limitations include high energy consumption, large equipment investment, operational complexity, thermal degradation of carotenoids during MAE, and enzyme dependency of EAE.

To address these challenges, Rezaee and colleagues pioneered dispersive liquid-liquid microextraction (DLLME) [28], a groundbreaking technique characterized by operational simplicity, rapid, and cost-effective with high extraction recoveries (ERs) and enrichment factors (EFs) [28–31]. The integration of deep eutectic solvents (DESs) further revolutionized DLLME by replacing toxic organic solvents. DESs, typically composed of hydrogen bond donors (HBDs) and hydrogen bond acceptors (HBAs), offer several benefits, including low toxicity, cost-efficient, ease of preparation, biocompatibility, and biodegradability [32,33]. Hydrophobic DESs (HDESs) are particularly effective for extracting lipophilic bioactive compounds. For instance, HDES combining DL-menthol [34] and lactic acid (HBD) has been used to extract lycopene from tomato pomace. Fan et al. (2022) developed a thymol-fenchol natural HDES (NADES) that enhanced microalgal lutein extraction efficiency sixfold compared to conventional n-heptane-ethanol-water systems and reduced processing time by 50 % [35,36]. In vitamin E extraction, Xie et al. [37]. developed an efficient method using DES-based liquid phase microextraction (DES-LPME) combined with reversed-phase high-performance liquid chromatography (RP-HPLC). These studies highlight HDES as a promising extractant for carotenoids and vitamin E from vegetables and fruits. However, previous research has predominantly focused on using single types of HDES to extract a limited range of nutrients, with no systematic investigation into the simultaneous extraction of both vitamin E and carotenoids.

This study aims to develop a novel HDES-based DLLME method coupled with HPLC for the simultaneous extraction and quantitation of vitamin E and carotenoids in fruits and vegetables. The Conductor-like Screening Model for Real Solvents (COSMO-RS) was employed to screen and identify optimal HDESs for this extraction process. The extraction conditions were systematically optimized to establish a robust HDES-DLLME method, coupled with an HPLC method utilizing a YMC C30 column for high-resolution separation and quantitation of carotenoids and vitamin E. The integrated HDES-DLLME/HPLC approach was then applied to quantify these nutrients in diverse vegetables and fruits, providing a deeper understanding of their compositional variability in plant-based foods. This research provides valuable insights for nutrition experts, supporting the selection of nutrient-dense plant-based foods and evidence-based dietary recommendations.

2. Material and methods

2.1. Chemical and materials

α-, γ-, δ-, β-Tocopherol, lutein, and β-carotene standards were purchased from Aladdin (Shanghai, China). Zeaxanthin and lycopene were bought from Macklin (Shanghai, China). β-Cryptoxanthin and α-carotene were obtained from Extrasynthese (Genay, France) and Santa Cruz Biotechnology (Santa Cruz, CA, USA), respectively. The internal standards *dl*-tocol and echinenone were obtained from Tama Biochemical (Tokyo, Japan) and CaroteNature (Lupsingen, Switzerland), respectively. Chromatographic grade methanol, isopropanol, hexane, and analytical grade methanol, ethanol, acetic acid, acetonitrile, acetone, ethyl acetate were all bought from Kermel (Tianjin, China). *DL*-menthol, propionic acid, butyric acid, hexanoic acid, heptanoic acid, caprylic acid, and decanoic acid were obtained from Aladdin (Shanghai, China). The vortex was obtained from Qiling beier (Haimen, China), the centrifuge was bought from Thermo Fisher (Hamburg, Germany), and the magnetic stirrer was obtained from Guohua HJ-3 (Jiangsu, China).

2.2. COSMO-RS calculation

The molecular structures of the HBDs, HBAs, and the vitamin E and carotenoid isoforms were optimized using Gaussian 09 software. The DFT-B3LYP-6–31 G (d, p) basis set was employed for the geometry optimization. The optimized structures were saved as *.log files and then converted to *.ml2 files for subsequent analysis. Next, the molecules were calculated using the BP-TZVP basis set in TURBOMOLE software, generating the *.cosmo files. Finally, these *.cosmo files were imported into COSMOThermX software to calculate the infinite dilution activity coefficients.

2.3. Preparation of fruit and vegetable samples

Fresh produce was procured from certified markets in Xi'an, Shaanxi Province (34°16′N, 108°54′E). Following USDA washing protocols, fruits were manually peeled, and vegetables that had non-edible portions (stems, seeds, damaged tissues) meticulously excised prior to processing. They were then cut into small pieces and homogenized using a homogenizer. The resulting homogenate was then transferred to centrifuge tubes and stored at −20 °C for subsequent analysis.

2.4. Preparation and characterization of DESs

DESs were prepared by mixing HBAs and HBDs in the specified molar ratios, as outlined in **Table S1**. The accurately weighed donors and acceptors were mixed in a 10 mL round-bottomed glass vial. The mixture was heated and stirred at 80 °C for 1 h using a thermostatic magnetic stirrer to ensure thorough mixing. Subsequently, both the heating and stirring were stopped, allowing the mixture to cool to room temperature, resulting in the formation of a clear, transparent, and homogeneous liquid. The successful formation of the menthol-thymol (1:2) DES was confirmed by Fourier transform-infrared spectroscopy (FT-IR, Nicolet iS50, Thermo Scientific, USA). The eutectic behavior and phase transition properties of the DES were characterized using differential scanning calorimetry (DSC, Discovery DSC250, TA Instruments, USA). For DSC analysis, samples were sealed in standard aluminum pans and subjected to a controlled heating program ranging from −90 °C to 80 °C at a heating rate of 10 K/min.

2.5. Extraction procedure

2 g sample was placed in a 10.0 mL centrifuge tube. To this, 10 μL of

methanol solution containing 0.01 mg/mL echinenone and 0.3 mg/mL *dl*-tocol was added as internal standards. Then, 1.1 mL of 2.1 M DES (extractant) in acetonitrile (dispersant) was added. After vortexed for 3 min, the solution was centrifuged at 10,000 rcf for 10 min at 4 °C. The resulting extract phase was collected for HPLC analysis.

2.6. Saponification

Saponification was performed only on samples containing lutein esters, chlorophyll, or those with high lipid and low carotenoid content [13,38]. After extraction, the supernatant was mixed with a methanol solution containing 0.1 % (w/v) butylated hydroxytoluene (BHT) and 10 % (w/v) potassium hydroxide (KOH) at a volume twice that of the extraction phase. The mixture was saponified under dark conditions at room temperature for 4 h, after which HPLC analysis was performed.

To evaluate the impact of the saponification pre- vs post-extraction, comparative experiments were conducted. Orange homogenate (2 g) was spiked with echinenone (0.01 mg/mL) and dl-tocol (0.3 mg/mL) in methanol. The saponification procedure involved 1000 μL of methanol with 0.1 % BHT (w/v) and 10 % KOH (w/v), reacting for 4 h at room temperature. For DLLME, 1100 μL of the 2.1 M menthol-thymol acetonitrile solution was used as the extractant. The extraction process involved vortex mixing for 3 min followed by centrifugation at 10,000 rcf for 10 min at 4 °C. Saponification was performed either before (pre-) or after (post-) the DLLME extraction, and the final solutions were adjusted to 1.5 mL for subsequent HPLC analysis.

2.7. Quantitation of carotenoids and vitamin E

The analysis of vitamin E and carotenoids was conducted using an LC-20AD HPLC system (Shimadzu, Kyoto, Japan). The system included a 20A3 degasser, two LC-20AD pumps, a SIL-20A autosampler, a CTO-20AC column oven, an LC-PDA detector, and Lab Solution software. A YMC C30 column (250 × 4.6 mm i.d., 5 μm) and a 10 × 4.0 mm i.d. C18 guard column were employed. The column temperature was set to 25 °C. The mobile phase consisted of methanol (A) and a 50 % (v/v) n-hexane/isopropanol mixture (B), following the gradient program: 0–3 min at 100 % A, 3–20 min transitioning from 100 % to 50 % A, 20–24 min from 50 % to 0 % A, 24–30 min at 0 % A, and 30–31 min transitioning from 0 % to 100 % A. The flow rate was maintained at 1.0 mL/min. The injection volume was 10 μL, and detection wavelengths were set at 290 nm for α-, γ-, δ-, β-tocopherol, 450 nm for lutein, zeaxanthin, β-cryptoxanthin, α-carotene, β-carotene, and 470 nm for lycopene. Molecules identification was achieved through comparing retention times and UV–vis spectra with standard substances.

2.8. Method validation

Method validation was conducted using diluted NFC apple juice (1:1 v/v with water) as the food matrix to simulate real sample conditions. Linearity, limits of detection (LODs) and quantitation (LOQs), and precision were assessed in this matrix. Carotenoids and vitamin E were quantified using internal standard methodology with echinenone and dl-Tocol as the internal standard, respectively. Matrix-matched calibration curves were constructed by spiking pre-prepared standard stock solutions (stored at −80 °C) individually into separate aliquots of the diluted apple juice matrix at five distinct concentration levels. Following individual extraction of each spiked sample, these matrix-matched standard solutions were analyzed in triplicate. Five-point matrix-matched calibration curves were established by plotting the peak area ratio (analyte-to-internal standard) against analyte concentration, using least-squares linear regression. LOD and LOQ were determined based on signal-to-noise ratios of 3:1 and 10:1, respectively. Method precision was evaluated through both intra-day ($n = 6$, same day) and inter-day ($n = 6$, over 6 consecutive days) experiments, where blank apple juice was spiked with analytes at low, medium, and high concentration levels. Method accuracy was assessed via recovery tests at three concentration levels across various vegetable and fruit samples. Recovery rates were calculated as the mean of three replicate measurements per fortification level.

2.9. Data processing and analysis

The experimental data were processed using EXCEL software, while graphs were generated using Origin software. Statistical analyses were carried out employing SPSS (version 18.0). A two-sample *t*-test was used to compare two groups. For comparisons involving multiple groups, an initial one-way analysis of variance (ANOVA) was performed. If a significant difference was identified, pairwise comparisons were executed using the LSD-*t*-test. Statistical significance was indicated by distinct lowercase letters (a, b, c, d, e, etc.) to denote differences between groups for the same indicator.

3. Results and discussion

3.1. Structural characteristics of target nutrients and DESs

As a statistical thermodynamic prediction method based on quantum chemical calculations, COSMO-RS has demonstrated effective in predicting the thermodynamic properties of various solvents [39]. It was therefore employed for the rational screening of high-efficiency DESs for vitamin E and carotenoid extraction. The surfaces colored according to the COSMO charge density σ, denoted the σ-surface, provide an intuitive tool for understanding the hydrogen bonding of both solutes and solvents. Additionally, the amount of surface area as a function of a specific surface screening charge density is illustrated by σ-profiles. In the σ-surface, areas with high positive charge density (HBD sites) are depicted in blue, where the surface screening charge density is negative and falls within the left region ($\sigma < -0.0084$ e·Å^2) in the σ-profiles, representing HBD capacity. Conversely, negative charge density (HBA sites) is shown in red, while low positive and negative charge densities (nonpolar sites) are displayed in green and yellow, corresponding to the right ($\sigma > +0.0084$ e·Å^2) and intermediate (-0.0084 e·Å$^2 < \sigma < +0.0084$ e·Å^2) regions in the σ-profiles, respectively. As illustrated in Fig. 1**(a) and (b)**, the target phytochemicals had numerous nonpolar surface sites, with σ-profiles predominantly located in the nonpolar region. Nonpolar carotenoids (lycopene and β-carotene) exhibited σ-profiles exclusively confined to the nonpolar region, consistent with their nonpolar hydrocarbon backbone lacking polar functional groups. Conversely, oxygenated compounds (α-tocopherol, lutein, zeaxanthin, and β-cryptoxanthin) demonstrated σ-profiles extending into both the HBD and HBA polar regions, manifesting blue and red regions on their surfaces. Among them, the σ-profile of α-tocopherol displayed a broader distribution in the HBD region, reflecting its HBD character due to its phenolic chromanol ring. In contrast, the σ-profiles of lutein, zeaxanthin, and cryptoxanthin exhibited a wider distribution in the HBA region, suggesting their HBA ability.

As shown in Fig. 1**(c)**, choline chloride (ChCl), L-carnitine (Lca) and betaine (Bet) exhibited high polarity, with their σ-profiles displaying more pronounced peaks and broad extension in the HBA region, indicating strong HBA ability. In contrast, thymol [36], menthol (Men) and lauric acid (Lau) showed strong distribution in the nonpolar region, with varying extension into the polar region. The σ-profiles of thymol and lauric acid exhibited significant extension into the HBD zone, while menthol displayed predominant extension into the HBA region. The classification of thymol and lauric acid as HBA or HBD in DESs depends on the relative hydrogen bond donor/acceptor strengths within the mixture. As shown in Fig. 1**(d)**, nonanoic acid (Non), octanoic acid (Oct), hexafluoroisopropanol (Hexa), and thymol [36] all featured high peaks in the nonpolar region with more pronounced distribution extending into the HBD region. Similarly, acetic acid (Ace), lactic acid (Lac), and pyruvic acid (Pyr) exhibited broader distributions in the HBD region, categorizing them as primary HBDs. Compounds like menthol,

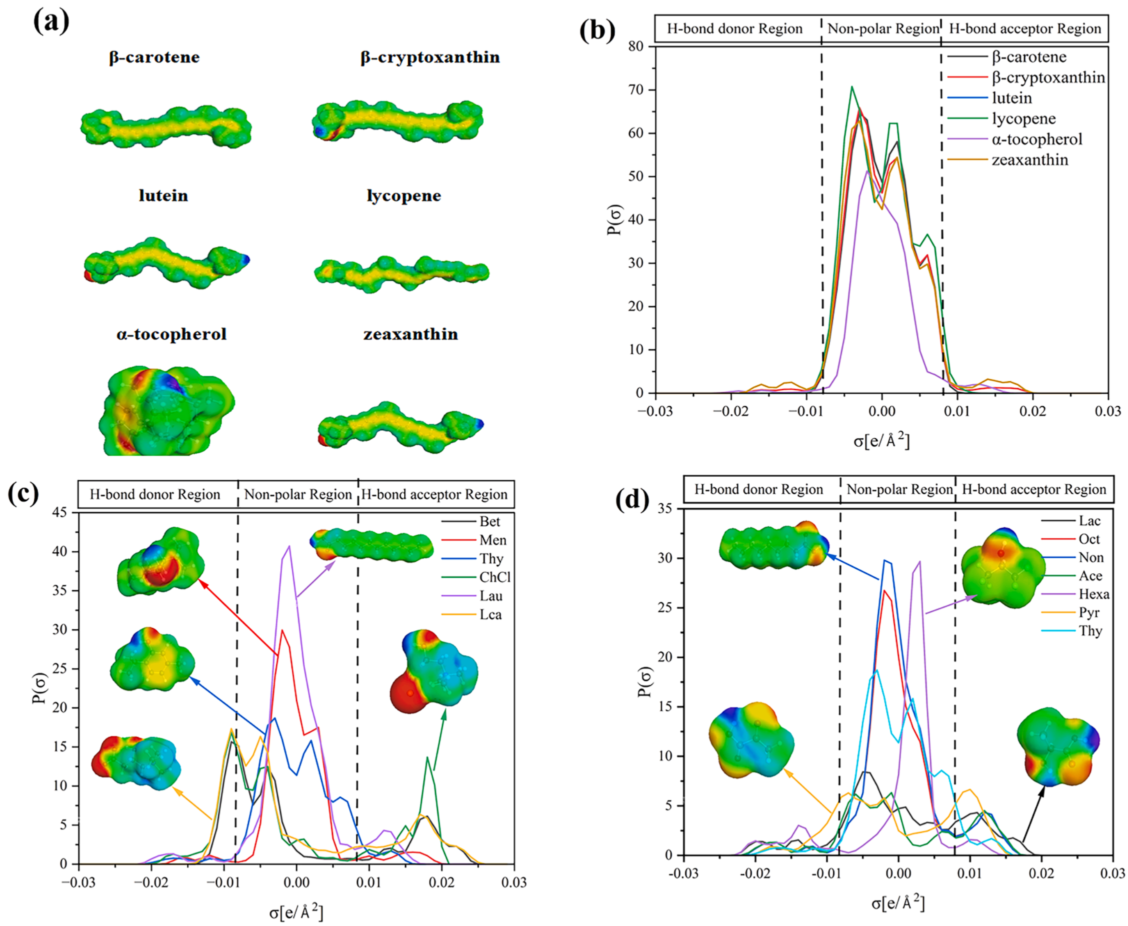

Fig. 1. σ-Surfaces and σ-profiles of target compounds and DES components. (a) σ-Surfaces of target compounds, (b) σ-Profiles of target compounds, (c) σ-Surfaces and σ-profiles of HBDs, (d) σ-Surfaces and σ-profiles of HBAs.

thymol, and long chain fatty acids, which show balanced distributions in both the polar and nonpolar regions, may serve as effective extractants for vitamin E and carotenoids due to their distribution in the nonpolar region.

3.2. DES screening by COSMO-RS

The selection of optimal DES for phytochemical extraction presents a significant challenge when relying on conventional trial-and-error approaches, which are resource-intensive, time-consuming, and require substantial reagents and manpower. To address this, we employed the COSMO-RS theory to systematically screen potential solvents. Initially, 34 candidate DESs were selected via a literature review encompassing common hydrophobic DESs based on choline chloride (ChCl), L-carnitine (Lca), betaine (Bet), thymol, menthol (Men), and long-chain fatty acids [40,41]. This selection aimed to cover a broad range of polarities and hydrogen-bonding capabilities. These candidates were first screened using COSMO-RS predictions to assess their theoretical affinity for the target nutrients. The key thermodynamic parameter for this evaluation was the solute's relative solubility at infinite dilution (Xi), calculated using Eq. (1) [42].

$$lgX_i = lg\{exp[(\mu_i^{pure} - \mu_i^{solvent}) / PT]\} \tag{1}$$

Where X_i represents the relative solubility of the solute i, μ_i^{pure} and $\mu_i^{solvent}$ are the chemical potentials of the pure compound i (the solute, i.e., carotenoids and vitamin E) and its infinitely diluted state in the solvent, respectively.

To account for the structural diversity among target analytes, five model compounds were selected: nonpolar carotenes (lycopene, β-carotene), polar xanthophylls (lutein, β-cryptoxanthin), and α-tocopherol (as a vitamin E homolog [8]). The solubility data (Fig. 2) revealed a consistent pattern across all tested compounds: DESs with enhanced nonpolar character (menthol-thymol systems) generally exhibited superior dissolution of the target phytochemicals, whereas highly polar DESs (betaine-, L-carnitine-, and choline chloride-based DESs) performed poorly, aligning with σ-profile data. Specifically, for lutein, except menthol-thymol, thymol-fatty acid and lauric acid-fatty acid DESs exhibited significantly higher solubility than menthol-fatty acid systems, likely due to the stronger HBA basicity of menthol enhancing hydrogen bonding with fatty acids and reducing DES solvation ability. Conversely, menthol-thymol DES demonstrated superior solubility for β-cryptoxanthin, β-carotene, lycopene, and α-tocopherol. The higher solubility of α-tocopherol in menthol-fatty acid DESs than thymol-fatty acid systems aligned with the HBD character of α-tocopherol and higher HBA capacity of menthol, enabling stronger hydrogen bonding. Overall, these results highlight the exceptional solubilizing power of menthol-thymol DESs for diverse bioactives, stemming from the balanced distribution of polar and nonpolar sites on their components, which allows interaction with a wide range of solute polarities. This systematic comparison underscores the critical role of DES formulation in extraction efficiency for structurally diverse phytochemicals.

3.3. Experimental screening of DES

Twelve DES candidates were selected based on COSMO-RS predictions to assess their efficiency in extracting carotenoids and vitamin E. As shown in Fig. 3**(a)**, the extraction efficiency, calculated as the

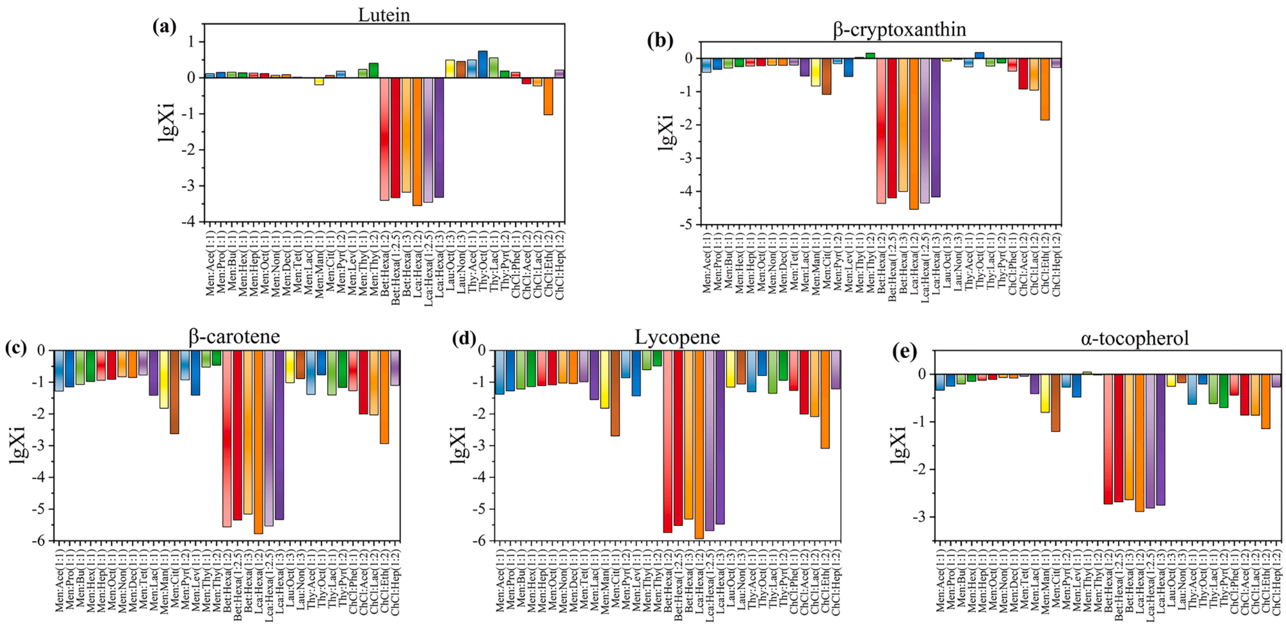

Fig. 2. The logarithm of the relative solubility of lutein (a), β-cryptoxanthin (b), β-carotene (c), lycopene (d), and α-tocopherol (e) in different types of DESs.

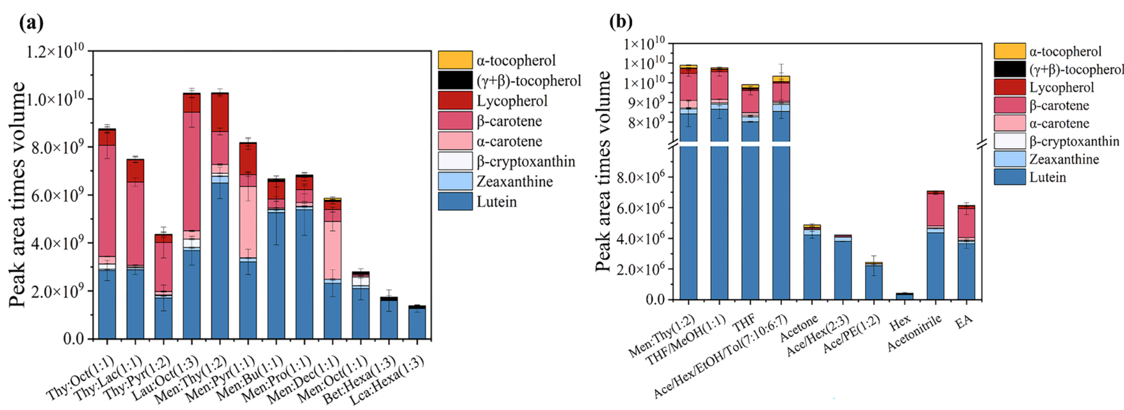

Fig. 3. The carotenoids and vitamin E extraction results using the selected DESs (a) and traditional organic solvents (b).

product of peak area and extract volume, demonstrated a strong correlation with the capacity trends predicted by COSMO-RS. Among the tested solvents, menthol-thymol (1:2) and lauric acid-octanoic acid (1:3) demonstrated the highest efficiency, followed by thymol-octanoic acid (1:1), and menthol-pyruvic acid (1:1). Conversely, menthol-octanoic acid (1:1) exhibited significantly reduced efficacy, while betaine-hexafluoroisopropanol (1:3) and L-carnitine-hexafluoroisopropanol (1:3) achieved the lowest yields. These experimental results confirmed the predicted extraction performance of the DESs.

The comparative extraction profiles of the menthol-thymol (1:2) and lauric acid-octanoic acid (1:3) DESs are summarized in Fig. S1. The menthol-thymol (1:2) system demonstrated superior efficiency for lutein, zeaxanthin, and lycopene, whereas the lauric acid-octanoic acid (1:3) DES proved more effective for β-carotene and β-cryptoxanthin. The extraction capacities for α-carotene, α-tocopherol, and ($\beta + \gamma$)-tocopherol were comparable between the two solvents, with menthol-thymol showing a slight advantage for α-carotene, while lauric acid-octanoic acid performed marginally better for α-tocopherol, ($\beta + \gamma$)-tocopherol. Notably, the total yields of fat-soluble nutrients were similar between the two systems. Based on this comprehensive comparison, the menthol-thymol (1:2) DES was selected as the optimal solvent. The primary rationale for this choice is twofold: first, it exhibited more consistent and overall superior extraction performance across the full spectrum of target analytes; second, its natural components (menthol and thymol) possess an established safety profile and offer broader applicability, particularly for potential use in sensitive sectors such as pharmaceuticals and food [43].

DES-based extraction demonstrated significant advantages over conventional organic solvents. While traditional solvent blends - including tetrahydrofuran (THF), THF/methanol (1:1 v/v), and acetone/hexane/ethanol/toluene (Ace/Hex/EtOH/Tol, 7:10:6:7, v/v) - showed relatively better efficiency among organic solvents, their extraction yields remained 1.2 % lower than the optimized Men: Thy (1:2) DES (Fig. 3**(b)**). Notably, single-component solvents (acetone, n-hexane (Hex), acetonitrile) and simple binary mixtures (acetone/n-hexane (Ace/Hex, 2:3), acetone/petroleum ether (Ace/PE, 1:2)) demonstrated substantially reduced efficacy. Therefore, the menthol-thymol DES represents a dual-action solution that merges benchmark extraction efficiency with sustainable processing advantages, outperforming all tested conventional solvents while addressing critical environmental limitations of organic extraction systems.

3.4. Structural characterization of the DESs

The successful formation and the key properties of menthol-thymol (1:2) DES used in this study were characterized by FT-IR spectroscopy and DSC. The FT-IR spectrum (Fig. 4**(a)**) revealed a significant shift of the hydroxyl stretching bands: the bands for pure menthol (~3370 cm^{-1}) and thymol (~3223 cm^{-1}) converged into a common, broader band centered around 3331 cm^{-1}. This observation indicates the disruption of the intrinsic hydrogen bonding in the pure components and the formation of a new intermolecular hydrogen-bonded network, which is characteristic of DES formation [44]. The presence of both components was further confirmed by characteristic peaks for aromatic C═C stretching (1618 and 1420 cm^{-1}), methyl C–H stretching (2956–2869 cm^{-1}), and phenolic C–O stretching (1227 cm^{-1}).

The DSC thermogram (Fig. 4**(b)**) displayed a single sharp endothermic peak at −56 °C. This melting point is significantly lower than those of the pure components (menthol, ~42 °C; thymol, ~50 °C), confirming a eutectic mixture with suppressed crystallinity. No separate melting transitions were observed, indicating homogeneity. Furthermore, the mixture remained in a stable, homogeneous liquid state at room temperature without phase separation over time, confirming its practical stability for extraction applications. Taken together, the spectroscopic, thermal, and physical data provide robust evidence for the formation of a homogeneous eutectic system, conclusively verifying the successful synthesis of the DES.

3.5. Optimization of DES-based DLLME conditions

3.5.1. Selection of the dispersant media

The dispersant plays a critical role in DLLME as it facilitates the formation of a microemulsion phase within the aqueous solution, enhancing mass transfer kinetics. Four potential solvents (acetonitrile, acetone, methanol, and ethyl acetate) were evaluated in this study. Methanol was excluded due to its failure to achieve complete phase separation with the target analytes. As illustrated in Fig. 5**(a)**, in general acetonitrile demonstrated superior extraction efficiency compared to the other candidates and was therefore selected for subsequent experiments.

3.5.2. Selection of the HBA/HBD ratios

The extraction performance of the menthol-thymol DES was systematically evaluated by varying its molar composition (1:1, 1:2, 1:3, and 1:4). While higher thymol ratios negatively impacted the extraction efficiency for β-carotene and lycopene (Fig. 5**(b)**), the 1:2 molar ratio achieved optimal extraction performance across all analytes and was selected for further investigation. This outcome is likely due to variations in intermolecular interactions within the DES components, which influence analyte solubility and partitioning behavior.

3.5.3. Selection of DESs concentration

The effect of DES concentration on extraction efficiency was investigated within the range of 1.9 M to 2.3 M. As depicted in Fig. 5**(c)**, extraction efficiency improved with increasing DES concentration from 1.9 M to 2.1 M, yielding the highest extraction efficiency for all analytes at 2.1 M. However, concentrations exceeding 2.1 M resulted in only marginal or even negative improvements, potentially due to increased viscosity at higher DES concentrations impeding mass transfer kinetics and complicating post-extraction handling. Therefore, 2.1 M was selected as the optimal concentration, balancing high extraction efficiency with manageable viscosity levels.

3.5.4. Optimization of the dispersed DES mixture volume

The impact of extractant volume on both analyte recovery and final concentration was assessed by testing volumes ranging from 800 μL to 1200 μL (Fig. 5**(d)**). While larger volumes generally improved analyte recovery, volumes greater than 1100 μL led to significant sample dilution, negatively affecting detection sensitivity. Thus, a volume of 1100 μL was determined to be optimal, maximizing analyte recovery while maintaining sufficient analyte concentration for reliable detection.

3.5.5. Selection of vortex time and centrifugation time

Vortex time (1–5 min) was found to exhibit negligible effects on most analytes' extraction efficiency, except for α-carotene, whose recovery increased with prolonged mixing time (Fig. 5**(e)**). A 3-min vortex mixing time was adopted as the optimal balance between ensuring complete phase homogenization and avoiding potential over-mixing. Finally, centrifugation time (5–30 min) was evaluated to determine the optimal duration for complete phase separation (Fig. 5**(f)**). Complete phase separation was consistently achieved within 10 min, with no further improvement in extraction efficiency observed at extended durations. A 10-min centrifugation protocol was thus established to optimize sample throughput without compromising separation efficiency.

Optimization identified the following conditions for the DES-based DLLME method using 1100 μL of 2.1 M menthol-thymol (1:2) DES in acetonitrile as the extractant, 3 min and 10 min as the suitable vortex mixing and centrifugation times, respectively. Compared to the conventional organic solvent extraction method, the DES-based DLLME method significantly simplified the pretreatment process by eliminating multiple extraction and centrifugation-concentration steps, reduced organic solvent consumption, improved extraction efficiency, shortened the pretreatment time, and enhanced the overall environmental friendliness of the process.

3.6. Optimization of saponification conditions

Key oxygenated carotenoids such as lutein, zeaxanthin, and β-cryptoxanthin, along with Vitamin E (tocopherols), often exist in both free

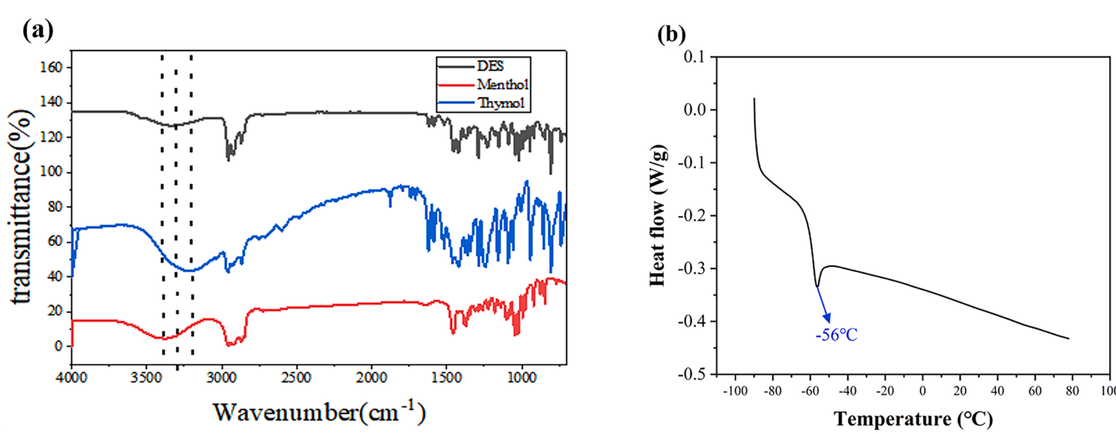

Fig. 4. Characterization of the menthol-thymol (1:2) DES. (a) FT-IR spectra, (b) DSC thermogram.

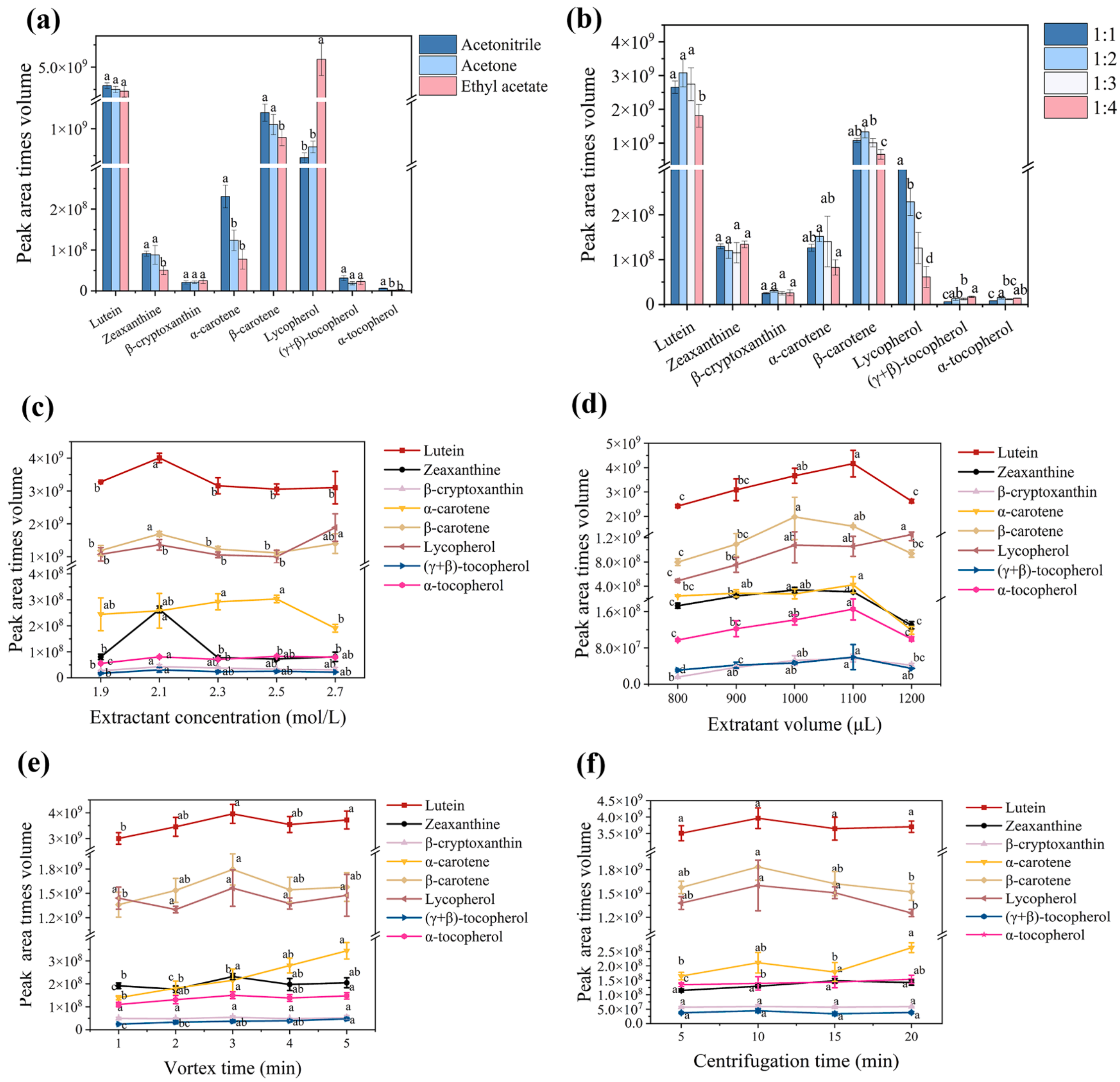

Fig. 5. Selection of extraction conditions. (a) Dispersant type, (b) HBA/HBD Ratio, (c) Extractant concentration, (d) Extractant volume, (e) Vortex time, and (f) Centrifugation time. Note: $n = 3$. Different lower-case letters (a-c) are significantly different ($P < 0.05$) as measured by the LSD-t-test.

and esterified forms within food matrices [45]. As the current HPLC method quantifies only the free forms, saponification is essential to hydrolyze esterified derivatives, enabling accurate total quantitation of these nutrient classes. Additionally, in samples with high concentrations of chlorophyll and co-extracted lipids, saponification serves to degrade these major interfering compounds, significantly reducing matrix effects and cleaning up the chromatographic profile [13,38]. Therefore, alkaline saponification was strategically implemented to achieve a dual purpose: converting esterified nutrients into their quantifiable free forms, and removing chlorophyll and saponifiable lipid interferences. This step is critical for obtaining a simplified chromatogram, accurate quantitation, and a holistic nutritional assessment.

3.6.1. *Comparison of saponification sequence (pre- vs. post-extraction)*

To integrate this essential step most effectively into the overall workflow, a systematic comparison was conducted to determine the optimal sequence—applying saponification either before the extraction procedure (pre-extraction) or after the initial extraction (post-extraction). Quantitative assessment via peak areas of target analytes (Table 1) revealed superior lutein and *β*-cryptoxanthin recovery when saponification was performed post-extraction ($P < 0.05$), consistent with Calvo's findings [46]. Conversely, zeaxanthin showed higher recovery in pre-extraction saponification, potentially reflecting

Table 1
Effect of saponification sequence ($n = 3$).

Peak area	lutein	zeaxanthin	*β*-cryptoxanthin
E-S	24,109^{a}	3773^{b}	431355^{a}
S-E	5531^{b}	6016^{a}	138088^{b}
Non-sp	2970^{c}	1711^{c}	71420^{c}

Note: E-S means saponification after extraction, S-E means saponification before extraction, Non-sp means non-saponification. Data followed by the different letters in the same column indicate significant different ($P < 0.05$).

differential stability during the alkaline treatment. After comprehensive consideration of the overall analytical performance, post-extraction saponification was chosen for further study to optimize the combined process. The increase in lutein, zeaxanthin, and *β*-cryptoxanthin recovery following post-extraction saponification confirmed the suitability of the DES system with the saponification mixture. The post-extraction saponification conditions were further investigated.

3.6.2. *Selection of KOH concentration*

The optimal KOH concentration for effective ester hydrolysis without excessive degradation was investigated under consistent experimental conditions. As shown in Fig. 6**(a)**, no statistically significant difference

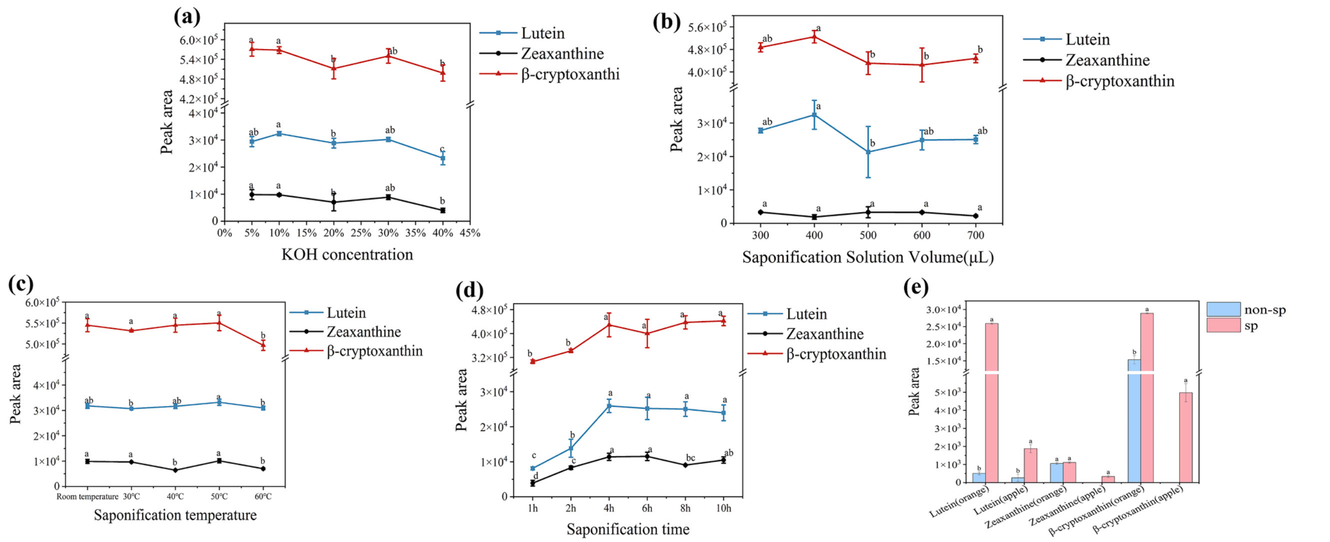

Fig. 6. Optimization of saponification conditions. (a) Selection of KOH concentration, (b) Selection of saponification solution volume, (c) Selection of saponification temperature, (d) Selection of saponification time, and (e) Comparison across different matrices. Note: $n = 3$. Data followed by the different letters indicate significant different ($P < 0.05$).

was observed in the concentration range of 5 %−10 % KOH ($P > 0.05$). However, concentrations exceeding 10 % induced noticeable carotenoid degradation, consistent with reported alkaline instability of these compounds. Conversely, a 10 % KOH concentration was chosen for saponification to balance complete ester hydrolysis with the preservation of analyte integrity.

3.6.3. Selection of saponification solution volume

The volume of the saponification solution affects both the reaction kinetics and the hydrolysis degree. As shown in Fig. 6**(b)**, at fixed extraction phase volume (200 μL), lutein and β-cryptoxanthin recovery peaked at a saponification solution volume of 400 μL (representing a 2:1 ratio to extraction phase volume). Volumes beyond 400 μL resulted in decreased recoveries, likely due to dilution effects affected the reaction. Zeaxanthin recovery showed volume-independent ($P > 0.05$). Thus, 400 μL of the saponification solution was determined to be optimal for maximizing lutein and β-cryptoxanthin recovery.

3.6.4. Selection of saponification temperature

Temperature optimization revealed its dual role in saponification efficiency and analyte stability. While a moderate increase in temperature accelerates the hydrolysis reaction rate by enhancing molecular movement, excessively high temperatures promote carotenoid degradation. Fig. 6**(c)** illustrated that peak areas (representing analyte concentration) decreased at higher temperatures, indicating a detrimental effect on analyte stability, aligning with reported carotenoid instability under thermal stress [47]. Room temperature (25 °C) was therefore selected as the optimal saponification temperature to ensure efficient hydrolysis while minimizing degradation.

3.6.5. Selection of saponification time

The duration of the saponification reaction was optimized to ensure complete ester hydrolysis. As shown in Fig. 6**(d)**, analyte concentrations plateaued after 4 h of saponification, indicating that ester bonds were sufficiently hydrolyzed. Extended durations (up to 6 h) showed no significant analyte degradation ($P > 0.05$), which may be attributed to BHT's protective effect. Therefore, 4 h was determined to be the optimal saponification time.

3.6.6. Evaluation of saponification necessity in different matrices

The necessity and effectiveness of saponification can vary significantly depending on the fruit matrix. Fig. 6**(e)** compares the carotenoid profiles of apples and oranges with and without saponification. Without saponification, zeaxanthin and β-cryptoxanthin were below the LOD in apples, likely due to the naturally low concentrations of their free forms. Saponification substantially increased the total amount of measurable carotenoids in both fruits by hydrolyzing esterified forms into their free counterparts. Oranges exhibited a far greater overall increase in measurable carotenoids after saponification (a 7.12-fold increase in total carotenoids compared to apples) directly correlating with their higher native content of carotenoid esters. Specifically, the measured levels of lutein and β-cryptoxanthin increased dramatically in oranges post-saponification (50.86-fold and 1.89-fold, respectively), whereas the increase for zeaxanthin was minimal (1.05-fold). This pattern reflects the distinct esterification profiles of carotenoids in oranges. In apples, a significant increase was observed primarily for lutein (7.15-fold).

Crucially, saponification significantly reduced chlorophyll interference in spinach, a high-chlorophyll matrix where overlapping chlorophyll peaks artificially suppressed lutein quantitation (**Fig. S2**). Post-saponification chromatograms showed significantly reduced or eliminated chlorophyll peaks, enabling accurate carotenoid detection. Conversely, in low-ester vegetables like tomatoes and carrots, or matrices with low chlorophyll content, saponification is often unnecessary due to minimal esterified carotenoids and reduced co-elution issues. This highlights the importance of considering the specific matrix composition when implementing saponification.

3.7. Development and validation of the HPLC method

This study employed chromatographic conditions optimized from our previous work on simultaneous determination of fat-soluble vitamins and carotenoids in human serum [48]. A C30 column was selected as the stationary phase to achieve baseline separation of lutein and zeaxanthin isomers. Vegetables and fruits contain carotene, the precursor of vitamin A (retinol), but not retinol itself. Consequently, retinol was not detected in this study. Similarly, 25-OH-VD3, a metabolic product of vitamin D used as an indicator for evaluating the nutritional status of vitamin D, was also not detected in this study. Except for α- and γ-tocopherol, which were the main forms of Vitamin E in serum, β- and δ-tocopherol were also detected in vegetables and fruits. Meanwhile, β- and γ-tocopherol co-eluted and were analyzed together. The mobile phase, composed of methanol and n-hexane/isopropanol, enabled separation through gradual elution as the proportion of n-hexane/isopropanol increased. The gradient elution condition was described in Section 2.7. All analytes were separated within 30-min (Fig. 7). The analytical specificity was confirmed through the use of specific absorption wavelengths, retention times and the standard addition method.

System suitability of the method was evaluated based on retention times, peak symmetry, resolution factors and theoretical plates. **Table S2** provides the corresponding chromatographic parameters at the concentrations of lutein 12.5 μg/mL, zeaxanthin 6.25 μg/mL, *β*-cryptoxanthin 6.25 μg/mL, *α*-carotene 12.5 μg/mL, *β*-carotene 14 μg/mL, lycopene 15 μg/mL, *δ*-tocopherol 12.5 μg/mL, (*β*+*γ*)-tocopherol 25 μg/mL, and *α*-tocopherol 67.5 μg/mL. Tailing factors for carotenes (ranging from 0.95 to 1.05, except *α*-carotene) were much better than those for Vitamin E, consistent with the column design for carotenoids. Most analytes achieved resolution factors larger than 1.5, indicating baseline separation, except for *β*- and *γ*-tocopherol, *α*-carotene and *β*-carotene. Additionally, an internal standard method was employed, using dl-Tocol for fat-soluble vitamins and echinenone for carotenoids. As previously reported, significant differences were observed in the slopes of the calibration curves generated in methanol without DLLME preparation versus after DLLME with standards dispersed in simulated serum. This indicated that the matrix effect correction provided by the internal standard, particularly for fat-soluble vitamins, was insufficient. To address this, in this study, diluted NFC apple juice (1:1, v/v with water) was used as the matrix for developing calibration curves after DES-based DLLME. The validation results were presented below.

The analytical method's performance was rigorously validated. Matrix matched quantitative analysis of vitamin E and carotenoids employed dl-tocol and echinenone as internal standards. Calibration curves were constructed by plotting the ratios of analyte-to-internal standard peak areas against concentration levels. Method sensitivity was assessed using limits of detection (LODs, defined as $S/N = 3$) and quantitation (LOQs, $S/N = 10$). As summarized in Table 2, all analytes exhibited excellent linearity ($R^2 \geq 0.9968$) across tested concentration ranges. LOQs ranged from 0.0302 to 0.0662 μg/mL for carotenoids, and 0.4717 to 0.6050 μg/mL for vitamin E.

3.7.1. Precision and accuracy

The precision of the method was evaluated using diluted NFC apple juice (1:1, v/v with water) as the matrix for spiking experiments. Intra-day precision ($n = 6$) yielded relative standard deviations (RSDs) ranging from 2.60 % to 16.38 %, and inter-day precision ($n = 6$) gave RSDs between 2.54 % and 16.18 %. These results demonstrate the method's satisfactory repeatability and precision (Table S3). To assess accuracy, recovery experiments were performed in three representative food matrices: pear, banana, and tomato. As shown in Table 3, the recovery rates (RRs) for the target analytes ranged from 80.04 % to 111.95 %, with associated RSDs between 0.36 % and 13.65 %. These findings confirm the accuracy and robustness of the method across diverse and complex food matrices. It is therefore considered suitable and reliable for the efficient enrichment and accurate quantitation of vitamin E and carotenoids in a wide variety of vegetables and fruits.

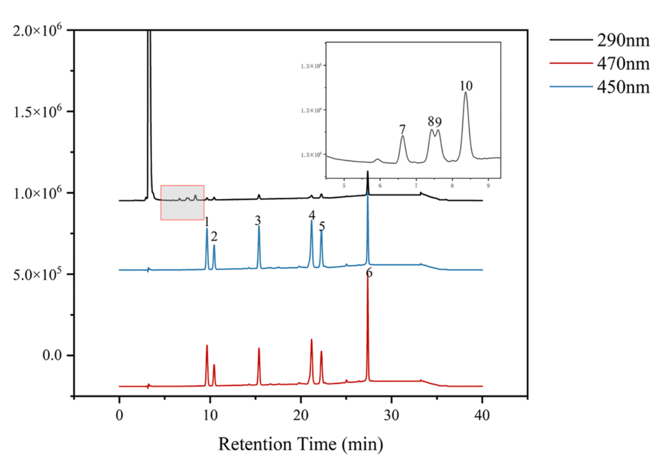

Fig. 7. HPLC chromatogram of a carotenoid and tocopherol standard mixture. Peak identification: 1, Lutein; 2, Zeaxanthin; 3, *β*-Cryptoxanthin; 4, *α*-Carotene; 5, *β*-Carotene; 6, Lycopene; 7, *δ*-Tocopherol; 8, *γ*-Tocopherol; 9, *β*-Tocopherol; 10, *α*-Tocopherol.

Table 2
Regression equation, correlation coefficient, linear range, detection limit, and quantitation limit of carotenoids and vitamin E.

Analyte	regression equation	R[2]	linear range (μg/mL)	LOD[a] (μg/mL)	LOQ[b] (μg/mL)
lutein	$y = 3.3119x - 0.1069$	0.9994	0.125–12.5	0.0128	0.0428
zeaxanthin	$y = 3.0589x + 0.2485$	0.9989	0.0625–6.25	0.0134	0.0448
β-cryptoxanthin	$y = 5.1265x + 0.2383$	0.9997	0.0625–6.25	0.0091	0.0302
α-carotene	$y = 4.4494x + 0.2429$	0.9999	0.125–12.5	0.0143	0.0471
β-carotene	$y = 2.5121x + 0.1727$	0.9993	0.14–14	0.0164	0.0545
lycopene	$y = 1.6955x + 0.0797$	0.9996	0.15–15	0.0199	0.0662
δ-tocopherol	$y = 0.2482x - 0.0078$	0.9989	0.125–12.5	0.1415	0.4717
(*β*+*γ*)-tocopherol	$y = 0.5066x + 0.0473$	0.9996	0.25–15	0.1816	0.6050
α-tocopherol	$y = 0.1685x + 0.1378$	0.9968	0.3375–67.5	0.1511	0.5037

[a] Limit of detection ($S/N = 3$). [b] Limit of quantitation ($S/N = 10$).

3.8. Comparison with other methods from literature

To comprehensively evaluate the performance of the current method, it was compared with conventional extraction techniques across key parameters: organic solvent consumption, sample preparation time, method recovery, and detection limit. The results, summarized in **Table S4**, demonstrated the advantages of DES-DLLME over conventional extraction methods in most performance metrics. Specifically, compared to traditional LLE [11,49–53], the present method demonstrates significant advantages: a drastic reduction in solvent consumption (1.1 mL vs. 7–30 mL), a substantially shorter sample preparation time (approximately 20 min vs. 3–5 h), and lower detection limits for key analytes such as lutein, β-carotene, and tocopherols. In summary, compared to methods reported in previous literature, the present approach offers distinct advantages in greenness (significantly reduced solvent use), analysis efficiency (shorter preparation time), and sensitivity (lower detection limits). However, certain aspects such as the long-term stability and batch-to-batch reproducibility of DESs, and the general applicability of the method to a wider range of sample matrices warrant further investigation for broader implementation.

3.9. Determination of target analytes in fruit and vegetable samples

The method applicability was assessed through triplicate analyses of diverse fruit and vegetable samples under optimized conditions (**Table S5**). The resulting carotenoid and vitamin E profiles generally aligned with established US nutritional databases (e.g., USDA Food Data Central) and literature values for similar commodities. Substantial carotenoid levels in spinach (lutein: 7263 μg/100 g, *β*-carotene: 5169 μg/100 g), significant lycopene content in cherry tomato (2044 μg/100 g), and elevated *α*-tocopherol levels in kiwifruit (973 μg/100 g) were

Table 3
Spiked recovery rates of vitamin E and carotenoids in different matrices.

Analyte	Added (μg)	Found (μg) (RR ± RSD,%, $n = 3$)		
		Pear	Banana	Tomato
Lutein	0	nd	nd	0.73 ± 0.03
	1	0.97 ± 0.03 (96.6 ± 2.73)	1.06 ± 0.09 (105.53 ± 9.31)	1.55 ± 0.13 (81.43 ± 13.33)
	2	1.97 ± 0.08 (98.19 ± 4.22)	1.98 ± 0.25 (99.17 ± 12.48)	2.49 ± 0.05 (87.85 ± 2.64)
	5	4.42 ± 0.14 (88.49 ± 2.83)	4.37 ± 0.02 (87.31 ± 0.36)	4.76 ± 0.27 (80.61 ± 5.38)
Zeaxanthin	0	nd	nd	0.06 ± 0.01
	0.5	0.40 ± 0.01 (80.37 ± 2.46)	0.56 ± 0.04 (111.95 ± 7.97)	0.51 ± 0.02 (90.15 ± 3.12)
	1	1.06 ± 0.08 (105.80 ± 7.85)	0.95 ± 0.10 (94.59 ± 10.25)	0.99 ± 0.06 (93.08 ± 6.25)
	2.5	2.32 ± 0.07 (92.82 ± 2.90)	2.22 ± 0.07 (88.98 ± 2.79)	2.18 ± 0.13 (84.71 ± 5.38)
β-cryptoxanthin	0	nd	0.01 ± 0.01	0.18 ± 0.03
	0.5	0.43 ± 0.01 (84.25 ± 0.27)	0.45 ± 0.04 (88.61 ± 8.23)	0.67 ± 0.04 (97.79 ± 7.18)
	1	0.93 ± 0.06 (92.15 ± 5.71)	0.90 ± 0.12 (88.90 ± 12.31)	1.17 ± 0.02 (92.51 ± 1.73)
	2.5	2.18 ± 0.04 (86.69 ± 1.54)	2.16 ± 0.06 (86.27 ± 2.58)	2.78 ± 0.28 (103.83 ± 11.28)
α-carotene	0	nd	nd	0.64 ± 0.04
	1	0.89 ± 0.04 (89.43 ± 3.91)	0.86 ± 0.08 (86.41 ± 8.29)	1.63 ± 0.08 (98.68 ± 7.87)
	2	1.89 ± 0.11 (94.3 ± 5.39)	1.77 ± 0.27 (88.27 ± 13.65)	2.27 ± 0.09 (81.27 ± 4.29)
	5	4.54 ± 0.10 (90.86 ± 2.07)	4.11 ± 0.14 (82.26 ± 2.76)	4.64 ± 0.39 (80.04 ± 7.84)
β-carotene	0	nd	nd	3.60 ± 0.18
	1.12	0.99 ± 0.06 (88.85 ± 5.67)	0.97 ± 0.10 (86.37 ± 9.05)	4.74 ± 0.08 (101.24 ± 6.79)
	2.24	2.02 ± 0.10 (90.31 ± 4,63)	1.94 ± 0.30 (86.65 ± 13.58)	5.40 ± 0.13 (80.39 ± 5.91)
	5.1	4.81 ± 0.12 (94.35 ± 2.41)	4.46 ± 0.16 (87.54 ± 3.07)	8.11 ± 0.54 (88.29 ± 10.59)
Lycopene	0	nd	nd	23.50 ± 0.78
	1	0.86 ± 0.07 (86.52 ± 6.94)	0.93 ± 0.09 (93.48 ± 9.37)	45.65 ± 2.12 (88.58 ± 8.47)[a]
	2	1.78 ± 0.05 (89.13 ± 2.36)	2.04 ± 0.18 (101.87 ± 9.21)	71.19 ± 9.27 (86.22 ± 2.72)[b]
	5	4.42 ± 0.05 (88.41 ± 1.09)	4.12 ± 0.26 (82.45 ± 5.27)	-
δ-tocopherol	0	0.25 ± 0.01	nd	nd
	1	1.10 ± 0.07 (84.24 ± 6.74)	0.90 ± 0.09 (90.38 ± 8.84)	0.85 ± 0.13 (85.27 ± 13.14)
	2	1.94 ± 0.07 (84.44 ± 3.58)	1.69 ± 0.17 (84.71 ± 8.48)	1.64 ± 0.18 (81.86 ± 9.18)
	4	3.66 ± 0.39 (85.08 ± 9.81)	3.68 ± 0.06 (92.10 ± 1.59)	3.25 ± 0.18 (81.16 ± 3.63)
(γ+β)-tocopherol	0	0.14 ± 0.04	nd	nd
	1	1.02 ± 0.05 (87.93 ± 4.68)	0.98 ± 0.09 (97.95 ± 8.59)	1.00 ± 0.08 (99.90 ± 8.36)
	2	1.75 ± 0.16 (80.79 ± 7.80)	1.93 ± 0.12 (96.58 ± 6.08)	1.77 ± 0.09 (88.37 ± 4.67)
	4	3.77 ± 0.27 (90.80 ± 6.63)	4.22 ± 0.29 (105.47 ± 7.32)	3.49 ± 0.17 (87.36 ± 3.36)
α-tocopherol	0	nd	nd	9.69 ± 0.20
	2.7	2.23 ± 0.22 (82.78 ± 8.24)	2.57 ± 0.27 (95.27 ± 9.96)	12.21 ± 0.06 (93.52 ± 2.27)
	5.4	4.41 ± 0.24 (81.76 ± 4.42)	5.55 ± 0.43 (102.09 ± 7.89)	14.55 ± 0.20 (90.12 ± 3.77)
	13.5	12.10 ± 1.25 (89.66 ± 9.27)	13.19 ± 1.12 (97.73 ± 8.28)	20.98 ± 0.94 (86.87 ± 7.28)

nd: Not detected.
[a] Spiking concentration: 25 μg/mL.
[b] Spiking concentration: 50 μg/mL.

observed, while banana, apple, and eggplant matrices showed comparatively lower concentrations of the target analytes. Analysis of the data revealed distinct patterns: kiwifruit, starfruit, small farmer mango, spinach, sweet bell pepper, and colored bell pepper were found to be rich sources of lutein; watermelon, small farmer mango, cherry tomato, and spinach contained relatively higher levels of carotenoids (including β-carotene); and kiwifruit, sweet orange, and tomato were abundant in vitamin E. Concentrations for most analytes were found within expected ranges, although some deviations occurred. Significant examples include lower levels of saponified lutein in spinach (7263 μg/100 g vs. ~12,000 μg/100 g USDA cation) and saponified β-carotene in red bell pepper (528 μg/100 g vs. ~1600 μg/100 g). Furthermore, comparison between saponified and unsaponified fractions indicated a notable increase in lutein content after saponification for summer squash, yellow/red bell peppers, watermelon, sweet orange, and spinach. These differences may be influenced by factors such as varietal genetics, cultivation conditions, maturity at harvest and sample preparation (saponification). Minor variations, such as the moderately elevated α-tocopherol concentrations observed in kiwifruit (973 μg/100 g) and spinach (1241 μg/100 g) are consistent with natural variability across different cultivars and growing regions. Overall, the data indicate that the phytonutrient profiles of the analyzed Chinese produce are fundamentally consistent with USDA data, with observed differences attributed to geographical origin, agricultural practices, and post-harvest handling, all falling within expected natural variation.

4. Conclusion

This study successfully integrated COSMO-RS theory with experimental validation to identify the optimal DES—specifically menthol-thymol (1:2)—for the extraction of carotenoids and vitamin E from complex food matrices. The developed DES-based DLLME method, coupled with HPLC, achieved the sensitive and simultaneous quantitation of ten analytes. The method demonstrated excellent analytical performance, characterized by high sensitivity (low LOD/LOQ), good linearity ($R^2 > 0.99$), satisfactory precision (RSD < 16.38 %), and high recovery rates (80.04–111.95 %). The strategic implementation of saponification significantly enhanced the quantitation of specific carotenoids (lutein, zeaxanthin, and β-cryptoxanthin) by hydrolyzing esterified forms and mitigating chlorophyll interference, leading to a more comprehensive nutritional assessment. Compared to conventional extraction methods, this integrated approach offers significant notable advantages, including higher extraction efficiency, reduced organic solvent consumption, improved environmental sustainability, and enhanced analytical sensitivity. In summary, the integrated strategy presented here—combining predictive computational solvent design (COSMO-RS) with an optimized DES-DLLME protocol—provides a robust, efficient, and reliable framework for the comprehensive analysis of key lipophilic nutrients in plant-based matrices. This work advances the toolkit available for food quality control and nutritional research.

CRediT authorship contribution statement

Mengting Cui: Writing – original draft, Visualization, Data curation, Conceptualization. **Qingpeng Gao:** Visualization, Formal analysis, Data curation, Conceptualization. **Jinling Bai:** Visualization, Data curation. **Mengting Fu:** Formal analysis. **Zhiyu Liu:** Methodology. **Zhixin Ouyang:** Methodology. **Liyun Kong:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Formal analysis, Conceptualization. **Le Ma:** Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.chroma.2025.466656.

Data availability

Data will be made available on request.

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